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Next item →

1. Which one of the following is true about relations?

1 / 1 point

☒ A relation can be represented by a matrix

☐ A relation cannot be represented by a matrix

☐ A relation can be represented by an undirected graph

☐ None of these answers

👍 Nice work

Correct. A relation can be represented by its adjacency matrix.

2. Let A be a set with $A = \{1, 2, 3\}$ and let $\mathcal{R}$ be a binary relation on A defined by $\mathcal{R} = \{(a, b) \in A \mid a \leq b\}$. Which one of the following matrices represents the relation $\mathcal{R}$?

1 / 1 point

☐ $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

☒ $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

☐ $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

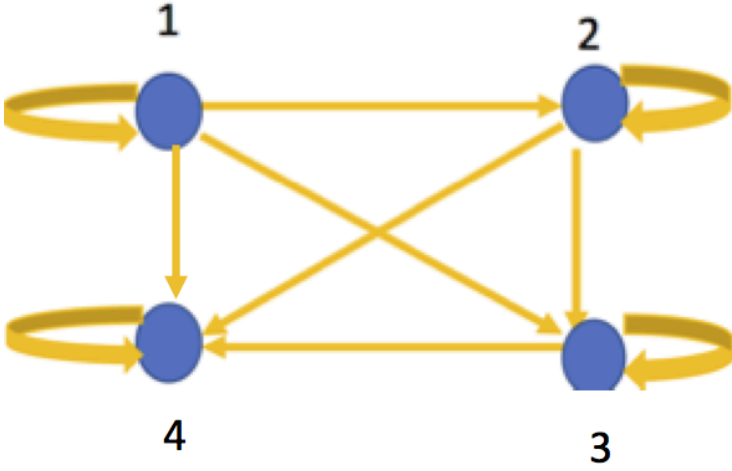
☐ $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

👍 Nice work

Correct. The matrix correctly represents the relation $\mathcal{R} = \{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$.

3. Let $A = \{1, 2, 3, 4\}$. Which one of the following relations on A is represented by the following digraph?

1 / 1 point



The relationship graph shown above have 4 vertices 1, 2, 3 and 4. It shows a loop at each vertex, an arrow from 1 to 2, an arrow from 1 to 3, an arrow from 1 to 4, an arrow from 2 to 3, an arrow from 2 to 4 and finally an arrow from 3 to 4.

☐ $\mathcal{R} = \{(a, b) \mid a, b \in A \text{ and } a \leq b\}$

☐ $\mathcal{R} = \{(a, b) \mid a, b \in A \text{ and } a \geq b\}$

☐ $\mathcal{R} = \{(a, b) \mid a, b \in A \text{ and } a > b\}$

☐ $\mathcal{R} = \{(a, b) \mid a, b \in A \text{ and } a < b\}$

👍 Nice work

Correct. The digraph correctly represents the relation $\mathcal{R} = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$.

4. Let $A = \{a, b, c\}$ and let $\mathcal{R}_1$ and $\mathcal{R}_2$ be two binary relations on A defined by their corresponding matrices $M_{\mathcal{R}_1}$ and $M_{\mathcal{R}_2}$ respectively, with:

1 / 1 point

$$M_{\mathcal{R}_1} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } M_{\mathcal{R}_2} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Which one of the following matrices represents the intersection of $\mathcal{R}_1$ and $\mathcal{R}_2$?

☐ $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

☒ $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

👍 Nice work

Correct. The intersection is represented by the logical AND (meet) of the two matrices.